

Project 4 — Milestone

Phase A

Project 4 milestone

Phase A in: [Project 4 — Designing Fair Comparisons](#)

What this checkpoint is for

This is the optional, ungraded Phase A worksheet (0 points) for Project 4. Treat it as preparation for the Phase B report and notebook. Answer using your assigned research question, and use vocabulary from [Module 4 Read 1](#) (study design, two-group tests/CIs, and confounding).

Phase A: Ten study-design questions

1. Variables. Identify the **explanatory** and **response** variables in your assigned question. For each, give an operational definition: exactly what you measure, units if numerical, and levels or cutoffs for categories (so another student could record data the same way).
2. Observational/Experimental units. What are the **people** or **items** from which data will be collected? Note: data from each of these items should put on the rows of the dataset with a column for each variable.
3. Means or proportions? Will you summarize the comparison with a **difference in means** (numerical response) or a **difference in proportions** (binary / two-category response)?
4. Hypotheses. State **null and alternative hypotheses** with symbols for the population quantities (e.g. μ_1 and μ_2 for means, or p_1 and p_2 for proportions). Define each symbol in a sentence in the context of the research question (e.g. μ_1 is the mean height of soy bean plants grown with fertilizer type A or p_1 is the proportion of radishes eaten by gophers in the winter).
5. RCT. Describe how you would run an ideal **randomized controlled trial** to answer the question, even if it is not feasible in real life. Be sure to mention random *assignment* of the groups in the explanatory variable along with when and what you would measure as the response variable.
6. RCT: two explanations. If the data from an RCT showed a difference between the two groups, give the two general types of explanations for that finding.
7. Observational study. Describe how an **observational study** could be carried out: who you observe, when, and what you record (note that in this design there is no random assignment of the explanatory groups).
8. Observational: three explanations. If the observational study showed a difference in responses between the two explanatory groups, give the three general types of explanations.
9. Confounding and DAG. Name one plausible **confounder** for the observational comparison. Draw a DAG with three nodes (explanatory, response, confounder) and arrows you could defend. In words, explain how the confounder is related to both the explanatory variable and the response.
10. Statistical test for “by chance” explanation. Using the [Methods table](#), name the appropriate test to address the “by chance” explanation for your variable types. State the sample-size / expected-count and other conditions you must check before trusting the test’s p -value.

What to turn in to Canvas (optional)

- Answers to the ten study-design questions above.

After the milestone

Complete Phase B on the [Project 4 page](#):

1. One reproducible R notebook (simulate + describe + check the confounder if applicable + test).
2. One-page study-plan report (PDF).